EART11300 Practical Skills Semester 2 (Chemistry 3)

Testing for phosphates: soil extraction and Palintest

Health and Safety

Hazard	Risk	Control Measures
Working with soil samples - exposure to contaminants (e.g. Pb^{2+}) and pathogens.	Ingestion, inhalation, skin contact pathways Pathogens – risk of infection and disease e.g. salmonella Heavy metal contaminants potentially toxic, carcinogenic.	 Work in spill trays. All soil spills or filtration residues disposed of in tiger bins. **Clean work surfaces and wash hands thoroughly after use.**
Solid sodium hydrogen carbonate ($NaHCO_3$) Effervescence upon reaction with acid.	Not classed as hazardous but extraction solution is mildly alkaline and causes eye irritation.	 Clean up spills immediately (solids with dustpan, solution with blue lab tissue – in both cases wipe benches and then wash hands)
Palintest tablets P1 	Causes severe skin burns and eye damage/irritation. May cause respiratory irritation. May damage fertility and the unborn child if swallowed.	 Place tablet directly into cuvette. Use forceps/dustpan for spills. Do not place the cap on the cuvette before the end of the colour development time (10 mins).
Palintest tablets P2 	Harmful if swallowed. Causes skin irritation. Causes serious eye damage/irritation. May damage fertility and the unborn child if swallowed.	 As above.
Glassware	Breakages lead to risk of cuts	Check apparatus for defects before use. Exercise care especially during clearing away/if glassware is wet. Tablet crusher must be applied gently to avoid breaking the cuvettes Follow correct disposal procedure for broken objects – demonstrators will use dustpan/forceps and place shards in red sharps bins.

By the end of this practical you should be able to:

1. Recall the steps required to perform a generic extraction for a soil sample as demonstrated by Olsen's method to extract available phosphate.
2. Be able to operate the Palintest spectrophotometric instrument independently including selection of analysis mode, preparation of the blank and preparation of sample to be analysed by the instrument using the correct protocol.

Background:

Testing soils for a variety of chemical species is an important aspect of environmental research and monitoring. In this practical the amount of available **phosphate (PO_4^{3-})** in soils will be measured using a similar protocol to that which is implemented in the field using the Palintest portable spectrophotometric instrument.

Phosphate is a source of phosphorus – an essential nutrient required for plant growth. Phosphate concentration is a marker for the available phosphate in soils which may correlate, for example, with the amount of fertiliser which has been applied to the soil to increase crop yields. Excessive application of phosphorus-rich fertilizers can lead to leaching of phosphate into local water sources with consequent risk of eutrophication in associated river systems.

Frameworks for the classification of soils according to their phosphorus content are not universal. In England, Wales and Northern Ireland, phosphorus content is measured against the Defra (Department for Environment Food and Rural Affairs) Index Scale[1] whereby phosphate is extracted from soil using **Olsen's method**[1],[2] and the amount quantified in mg/l.

Index	Concentration of phosphate via Olsen's method /mg/l
0	0-9
1	10-15
2	16-25
3	26-45
4	46-70
5	71-100
6	101-140
7	141-200
8	201 – 280
9	280

Target Indices are recommended to ensure optimal quantities of nutrients are available to promote crop yields whilst minimising excesses which could lead to high levels of contamination in associated water sources. Index 2 is the optimal concentration for arable, grassland and forage crops but Index 3 is ideal for vegetable crops.

The Olsen's method[1],[2] extraction solution is 0.5M sodium bicarbonate at pH 8.5. The measurement obtained correlates positively with increasing pH, extraction time, agitation of the extraction vessel, and temperature. The method presented today is sufficient to understand the principle of the extraction and to obtain some measurements, but consideration should be given to how changing the aforementioned factors might lead to a different result.

The concentration of phosphate in the extraction solution is determined by a **spectrophotometric** ("photometry") method using the Palintest instrument which will be familiar to you from the Mam Tor fieldwork in semester 1.

Aim: To measure the concentration of phosphate in soil samples and assess against the Defra phosphorous index classification

Theory (the chemistry!):

Extraction

The Olsen method[1,2] can be used for mildly acidic to alkaline/calcareous soils. Addition of 0.5M $NaHCO_3$ at pH = 8.5 increases the solubility of phosphates that are present in the soil as calcium/magnesium/aluminium/iron phosphates.
For example:

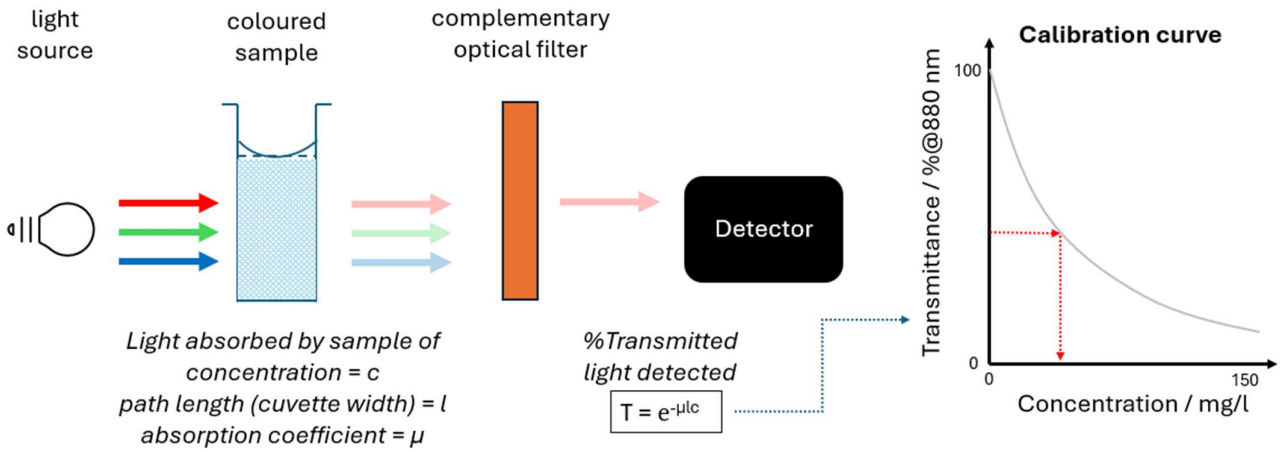
$$Ca_3PO_4(s) + 3NaHCO_3(aq) \xrightarrow{0.5M\ NaHCO_3\ pH=8.5} Na^+[H_2PO_4]^-(aq) + 2Na^+[HPO_4]^{2-}(aq) + 3CaCO_3(s)$$

The reactions precipitate carbonates and hydroxides whilst the phosphates remain in solution.

Spectrophotometry (aka photometry / colorimetry)

Colour reagents are added to form coloured complexes with species in the sample. In the case of the phosphate test, a transition metal compound (ammonium molybdenate) is added to form a blue complex with phosphate.

Complexes of transition metal ions are often intensely coloured. Interactions between pairs of electrons on ligands and the d-orbitals of the transition metal ion cause splitting of the d-orbitals energies into two sets separated by an energy gap. The energy gap corresponds to the energy of photons in the visible region of the EM spectrum. In the photometry instrument, the sample is exposed to a light source. Photons with the same energy as the sample-complex energy gap are absorbed whilst the remaining light is transmitted and detected by the instrument. The %transmitted light is related to the concentration of the sample-complex by the Beer-Lambert Law. A calibration curve is used to find the concentration of the species of interest from the transmittance:



In today's practical, the blue complex is analysed by the Palin test instrument using a complementary red filter and detection of light with wavelength =650 nm.

Procedure (aim to analyse two soil samples)

Part 1: Extraction

Note: the soils have already been **milled and sieved** (≤ 2.0 mm) to conserve sample representativeness and **dried** in preparation for phosphate extraction (although in the field Palintest tablets can be carried out on fresh, damp soil samples).

1. In the bottle provided, measure out 2.1g of solid sodium hydrogen carbonate ($NaHCO_3$) using the top-pan balance.
2. Using a measuring cylinder, add 50 ml deionised water to the extraction bottle and swirl/stir to ensure the sodium bicarbonate has completely dissolved.
3. Check the pH of the extraction solution is ~8.5 using a Pasteur pipette to drop a sample on to universal indicator paper.
4. Select one of the soil samples from the bulk samples transfer 2 ml of soil to the extraction bottle by taking a level measure with the 2ml scoop provided (be careful to prevent cross contamination).
5. Ensure the lid is tightly on the bottle and shake the mixture of soil and extraction solution for several seconds then check for effervescence which may occur with acidic soils. Release pressure by opening the lid if required. Repeat until total shaking time is **one minute**.
6. Set up gravity filtration apparatus (conical flask, filter funnel and Whatman No.1 filter paper). The filter paper should be folded into quartered or fluted form (fluted is more efficient).
7. Carefully pour the extraction mixture through the filter paper. The first few drops of filtrate may be 'turbid' (particularly cloudy with sediment) and if so should be discarded. Wait for the extraction filtrate to accumulate in the conical flask – be patient, this may take time!

Part 2: Measuring phosphate concentration using the Palintest instrument

a: Preparation of the 'blank'

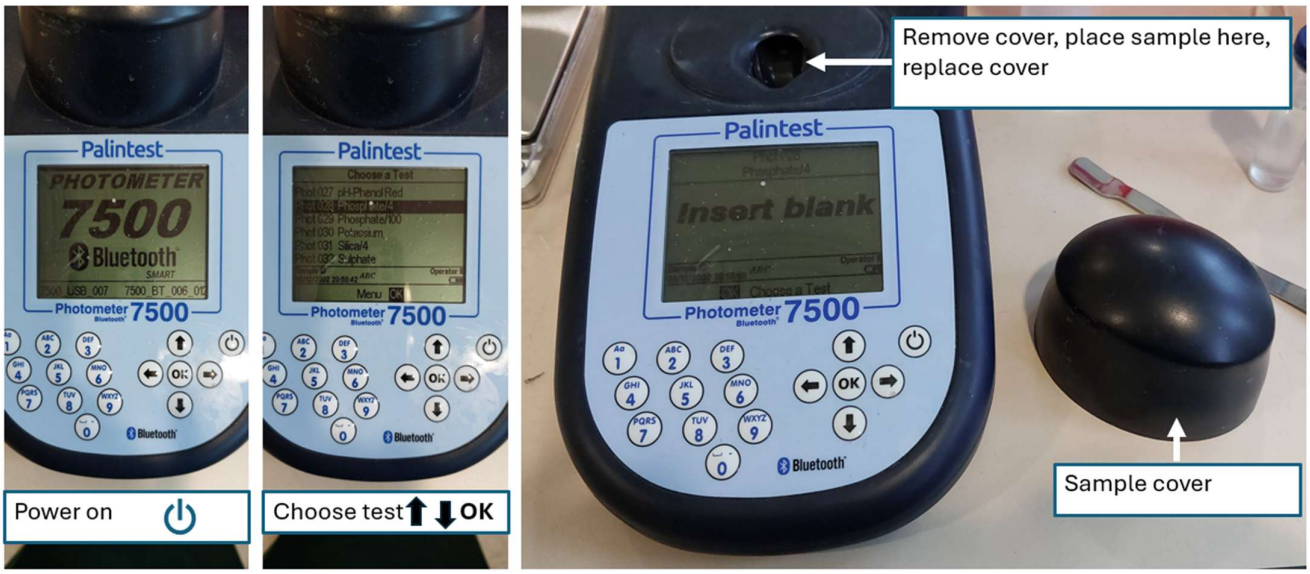
1. Using a clean 2ml syringe, carefully extract **exactly 2 ml** of extraction filtrate and add to a clean 10 ml plastic cuvette. Avoid touching the sides of the cuvette as any marks left might affect the reading on the photometer.
2. Use deionised water to dilute the filtrate and make up to the 10 ml mark on the cuvette (ensure the bottom of the meniscus rests on the white calibration line marked on the cuvette – eyes should be level with the calibration line when judging this).
3. Place the blue plastic cap on the cuvette.
   This is the 'blank' for the spectrophotometric method – do not add any further reagents to this cuvette.

b: Preparation of the sample to be analysed

4. Repeat steps 1 and 2 from part a) above to prepare a second cuvette containing diluted extraction filtrate.
5. Add one Palintest P1 tablet **directly** (i.e. push the tablet from its packaging straight into the cuvette, not onto your hand first) to this second cuvette. Use a clean plastic tablet crusher to press down on the tablet *gently* to speed up dissolution. The addition of this tablet might be accompanied by rapid effervescence – be aware in case any solution spills over the top of your cuvette (and try to avoid this by crushing gently!).

6. Add one Palintest P2 tablet and crush as required. **DO NOT place the cap on the cuvette** yet as continued effervescence may cause a build-up of pressure under the cap and may launch the cap off the cuvette**.**
7. Wait **10 mins** for the blue colour to develop.
8. Place the blue cap on the cuvette.

c: the Palintest spectrophotometric method



1. Turn the instrument on.
2. Select the correct test mode for this investigation: **<u>Phot 028 Phosphate/4</u>** by using the up/down arrows and pressing 'OK'.
3. When the display reads 'Insert blank', remove the dome-shaped sample cover, place the cuvette containing the **blank** into the instrument and replace the cover to shield the measurement from ambient light. Press 'OK' and wait until the display reads 'Insert Sample'.
4. When the display reads 'Insert sample', remove the blank, insert the sample cuvette and replace the instrument cover, press 'OK'. Wait for the instrument to display the result.
5. Read the instrument measurement for the sample and then multiply by x5 to account for the dilution carried out in the preparation of the blank/sample cuvettes. Record the concentration for the undiluted extraction filtrate in the results table below.
6. If the result indicates saturation (i.e. result is greater than range of instrument setting and the display reads '**>>**') you will need to prepare a further dilution of the sample and repeat the test (i.e. prepare new blank and coloured samples). Try preparing blank and sample using 1ml of extraction solution made up to the cuvette calibration line. You will then need to multiply the concentration result by factor 10 to obtain your result. Note: in the field, use the Palintest dilution tubes and syringe provided with the instrument to prepare diluted samples.

Compare results with your peers to obtain a range of measurements from which you can calculate a mean average.

Results Table

Soil sample	**Concentration measurement reported by Palintest mg/l**					**Classification (Defra Index)**
	1	2	3	Range	Average	
1						
2						
3						
4						
5						

Data Analysis

1. Using your results and results of some of your peers, complete the results table above to determine the range and mean average results of the soil samples.
2. Discuss the reasons for the range of results with reference to the factors which may affect the extraction process (see background /theory sections).

Evaluation of potential error sources

1. Why is it important to ensure the colour development tablets are transferred directly to the cuvettes? What effect would it have on your result if the tablets were not directly transferred e.g. if they were broken/transferred to the hand first?
2. Why should you not touch the sides of the cuvette? What effect might touching the sides of the cuvette have on your result?
3. Why should the particularly cloudy (turbid) first drops of the extraction filtrate be discarded? What effect would it have on your result if these were not discarded?
4. Why must the cuvettes, cuvette lids and tablet crushing rods be completely clean before you prepare your samples for photometry?

References and further information

[**1**] https://ahdb.org.uk/rb209 The Agriculture and Horticulture Development Board (AHDB an ALB of Defra) Nutrient Management Guide (RB209) – Section 1: Principles of nutrient management and fertiliser use pp 24 - 29

[**2**] S. Olsen & L. Sommers. Phosphorus. In: Methods of soil analysis; chemical and microbiological properties. Part 2. Ed. by A.L. Page, R.H. Miller and D.R. Keeney. American Society of Agronomy/Soil Science Society of America: Madison, Wisconsin, 403-430 (1982)